

**Linear Algebra (E)**

**2025 Fall**

**Final Exam**

**2026.1.11**

**Exam Duration: 3 Hours**

**Name:** \_\_\_\_\_

**Student ID:** \_\_\_\_\_

This exam includes 18 pages (including this page) and 6 problems. Please check to see if there is any missing page, and then write down your name and student ID number on this page and the first page of your answer sheets. Also write down the initials of your name on the top of every page of your answer sheets, in case they are scattered.

This exam allows one **SINGLE** cheat sheet (size A4, double-sided is fine). No calculator or any electronic device is allowed. Plagiarism of all kinds is strictly forbidden and will be severely punished.

Please write down your answers to the problems in the **SEPARATE ANSWER SHEETS**, and follow the following rules:

- **Always explain your answer.** You should always explain your answers. At least write down one sentence to explain your thoughts.
- **Write clearly and legibly.** Make sure that your writings can be read. The graders are NOT responsible to decipher illegible writings.
- **Partial credits will be given.**
- Blank spaces are provided in the exams. Feel free to use them as scratch papers. However, your formal answer has to be written in the **SEPARATE ANSWER SHEETS**, as required by the University.
- The total score of the exam is 62. No overflow.

| Problem | Points | Score |
|---------|--------|-------|
| 1       | 6      |       |
| 2       | 10     |       |
| 3       | 11     |       |
| 4       | 14     |       |
| 5       | 12     |       |
| 6       | 9      |       |
| Total:  | 62     |       |

**1. (Basic Problems)**

These problems are basic.

- (a) (2 points) Suppose a  $3 \times 3$  real matrix  $A$  has eigenvalues  $\lambda_1 = -1$ ,  $\lambda_2 = 1$ , and  $\lambda_3 = 2$ .
- Find the determinant of  $A$ .
  - Find the trace of the matrix  $A^2$ .

- (b) (2 points) Let  $B$  be a  $3 \times 3$  matrix with  $\det(B) = 4$ . Calculate the value of  $\det(2B^{-1})$ .

- (c) (2 points) Find the value of  $k$  that makes the following three vectors in  $\mathbb{R}^3$  linearly dependent:

$$\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 0 \\ k \\ 1 \end{bmatrix}.$$

**Answer:**1. Eigenvalues:

- Determinant is product of eigenvalues:  $\det(A) = (-1)(1)(2) = -2$ .
- Trace is sum of eigenvalues. Eigenvalues of  $A^2$  are  $\lambda_i^2 \in \{1, 1, 4\}$ .  $\text{trace}(A^2) = 1 + 1 + 4 = 6$ .

2. Determinant Properties:  $\det(2B^{-1}) = 2^3 \det(B^{-1}) = 8 \cdot \frac{1}{\det(B)} = 8 \cdot \frac{1}{4} = 2$ .3. Linear Dependence: Vectors are dependent if  $\det[\mathbf{u}, \mathbf{v}, \mathbf{w}] = 0$ .

$$\det \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & k \\ 1 & 0 & 1 \end{bmatrix} = 1(1 - 0) - 1(0 - k) + 0 = 1 + k.$$

$$1 + k = 0 \implies k = -1.$$

**2. (The Variable Parameter)**

Let  $C_x$  be a  $4 \times 4$  matrix depending on a real scalar parameter  $x$ :

$$C_x = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 2 & 0 & 0 \\ 0 & 1 & 1 & x \\ 1 & 1 & -x & 1 \end{bmatrix}.$$

- (a) (2 points) Without expanding the full determinant, find the coefficient of the  $x^2$  term in  $\det(C_x)$ .

- (b) (3 points) Using cofactor expansion or any method, show that the characteristic polynomial of  $C_x$  is:

$$p(\lambda) = (\lambda - 2)(\lambda - 1)(\lambda^2 - 2\lambda + x^2).$$

For what real values of  $x$  is the matrix  $C_x$  singular?

- (c) (3 points) Let  $x = 2$ . Find all 4 eigenvalues of  $C_2$ . Find a complex eigenvector corresponding to the eigenvalue with a positive imaginary part.

- (d) (2 points) Let  $M = (2I - C_x)(I - C_x)$  and  $N = C_x^2 - 2C_x + x^2I$ . Compute the matrix product  $MN$ . (Hint: Do NOT expand the matrices element-wise. Use the result from part (b)).

**Answer:**1. Coefficient of  $x^2$ :

The only way to get non-zero term in the big formula for determinant is to pick the (1,1) entry and (2,2) entry. Thus Coefficient is **2**.

2. Characteristic Polynomial:

Expand  $\det(C_x - \lambda I)$  along Row 2 (which contains only one non-zero entry,  $2 - \lambda$ ):

$$\det(C_x - \lambda I) = (2 - \lambda) \det \begin{bmatrix} 1 - \lambda & 0 & 1 \\ 0 & 1 - \lambda & x \\ 1 & -x & 1 - \lambda \end{bmatrix}.$$

Now, expand the  $3 \times 3$  determinant along Row 1:

$$\begin{aligned} \text{Determinant portion} &= (1 - \lambda) \det \begin{bmatrix} 1 - \lambda & x \\ -x & 1 - \lambda \end{bmatrix} + 1 \det \begin{bmatrix} 0 & 1 - \lambda \\ 1 & -x \end{bmatrix} \\ &= (1 - \lambda) [(1 - \lambda)^2 - (-x)(x)] + 1 [0 - (1 - \lambda)] \\ &= (1 - \lambda) [(1 - \lambda)^2 + x^2] - (1 - \lambda) \\ &= (1 - \lambda) [(1 - \lambda)^2 + x^2 - 1]. \end{aligned}$$

Simplifying the term in the brackets:

$$(1 - 2\lambda + \lambda^2) + x^2 - 1 = \lambda^2 - 2\lambda + x^2.$$

Combining this with the pre-factor  $(2 - \lambda)$ :

$$p(\lambda) = (2 - \lambda)(1 - \lambda)(\lambda^2 - 2\lambda + x^2) = (\lambda - 2)(\lambda - 1)(\lambda^2 - 2\lambda + x^2).$$

**Singular:** The matrix is singular when  $p(0) = 0$ :

$$p(0) = (-2)(-1)(x^2) = 2x^2.$$

Thus, it is singular if and only if  $x = 0$ .

3. Eigenvalues and Vector ( $x = 2$ ):

Roots of  $\lambda^2 - 2\lambda + 4 = 0$  are  $\lambda = \frac{2 \pm \sqrt{4 - 16}}{2} = 1 \pm i\sqrt{3}$ . Eigenvalues:  $2, 1, 1 + i\sqrt{3}, 1 - i\sqrt{3}$ .

**Eigenvector for  $\lambda = 1 + i\sqrt{3}$ :** Solve  $(C_2 - \lambda I)\mathbf{v} = 0$ . Row 2 implies  $(2 - \lambda)v_2 = 0 \implies v_2 = 0$ . The system reduces to: 1)  $(1 - \lambda)v_1 + v_4 = 0 \implies v_4 = (\lambda - 1)v_1 = i\sqrt{3}v_1$ . 3)  $v_2 + (1 - \lambda)v_3 + 2v_4 = 0 \implies -i\sqrt{3}v_3 + 2(i\sqrt{3}v_1) =$

$$0 \implies v_3 = 2v_1. \text{ Set } v_1 = 1. \mathbf{v} = \begin{bmatrix} 1 \\ 0 \\ 2 \\ i\sqrt{3} \end{bmatrix}.$$

4. Cayley-Hamilton:

The characteristic polynomial is  $p(\lambda) = (\lambda - 2)(\lambda - 1)(\lambda^2 - 2\lambda + x^2)$ . Note that  $M = (2I - C_x)(I - C_x) = (C_x - 2I)(C_x - I)$  (signs cancel). And  $N = C_x^2 - 2C_x + x^2I$ . Thus  $MN = p(C_x)$ . By the Cayley-Hamilton theorem,  $p(C_x) = \mathbf{0}$  (the zero matrix).

3. Consider the  $5 \times 5$  matrix  $A$ , which has a symmetric pattern resembling an hourglass:

$$A = \begin{bmatrix} a & 0 & 0 & 0 & b \\ 0 & a & 0 & b & 0 \\ 0 & 0 & a+b & 0 & 0 \\ 0 & b & 0 & a & 0 \\ b & 0 & 0 & 0 & a \end{bmatrix}.$$

- (a) (4 points) For generic  $a, b$ , find the spectral decomposition of  $A$ . That is, find an orthogonal matrix  $Q$  and a diagonal matrix  $\Lambda$  such that  $A = Q\Lambda Q^T$ .
- (b) (2 points) For what conditions of  $a, b$  will this matrix be indefinite?
- (c) (3 points) If  $a = b \neq 0$ , find a basis for the vector space  $V = \{p(A) : p \text{ is any polynomial}\}$ . (Remember to prove spanning and linear independence for your basis elements.)
- (d) (2 points) For  $a = b = 1$ , find a polynomial  $p(x)$  of degree 1 such that  $(A + I)^{-1} = p(A)$ .

**Answer:**1. Spectral Decomposition:

The matrix  $A$  acts on the standard basis vectors. By reordering the basis as  $\{e_1, e_5, e_2, e_4, e_3\}$ , the matrix becomes block diagonal with two  $2 \times 2$  blocks  $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$  and one  $1 \times 1$  block  $[a + b]$ . The eigenvalues of  $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$  are  $a + b$  (eigenvector  $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ ) and  $a - b$  (eigenvector  $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ ). The middle element corresponds to eigenvector  $e_3$  with eigenvalue  $a + b$ .

**Eigenvalues:**  $\lambda_1 = a + b$  (multiplicity 3),  $\lambda_2 = a - b$  (multiplicity 2).

**Orthonormal Eigenvectors (Columns of  $Q$ ):** For  $\lambda = a + b$ :

$$\mathbf{q}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{q}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{q}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}.$$

For  $\lambda = a - b$ :

$$\mathbf{q}_4 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ -1 \end{bmatrix}, \quad \mathbf{q}_5 = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}.$$

Thus:

$$\Lambda = \text{diag}(a + b, a + b, a + b, a - b, a - b), \quad Q = [\mathbf{q}_1 \quad \mathbf{q}_2 \quad \mathbf{q}_3 \quad \mathbf{q}_4 \quad \mathbf{q}_5].$$

2. Indefiniteness:

A symmetric matrix is indefinite if it has at least one positive eigenvalue and at least one negative eigenvalue. The eigenvalues are  $a + b$  and  $a - b$ . We require  $(a + b)(a - b) < 0$ .

$$a^2 - b^2 < 0 \implies |a| < |b|.$$

3. Note that  $A^2 = 2aA$ . This means  $V$  is two dimensional and a basis is  $I, A$ , since all higher order polynomials can use  $A^2 = 2aA$  to reduce to a degree one polynomial.

For the linear independence, the proof is also very easy. If  $sA + tI = 0$ , then the lower left entry suggests that  $s = 0$ , and then the upper left entry suggests that  $t = 0$ .



4. Polynomial Inverse:

For  $a = b = 1$ , we have  $A^2 = 2A$ . We want  $p(A) = c_1A + c_0I$  such that  $(A + I)(c_1A + c_0I) = I$ .

$$c_1A^2 + (c_0 + c_1)A + c_0I = I.$$

Substituting  $A^2 = 2A$ :

$$(2c_1 + c_0 + c_1)A + c_0I = I \implies (3c_1 + c_0)A + c_0I = I.$$

Comparing coefficients (since  $A$  and  $I$  are linearly independent here):

$$\begin{cases} c_0 = 1 \\ 3c_1 + c_0 = 0 \end{cases} \implies c_1 = -1/3$$

Thus, the polynomial is:

$$p(x) = -\frac{1}{3}x + 1.$$

4. We are trying to recover a signal  $\mathbf{x} \in \mathbb{R}^2$  from a measurement  $\mathbf{b} \in \mathbb{R}^3$ . The measurement process is modeled by  $A\mathbf{x} = \mathbf{b}$ , where:

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 3 \\ 2 \\ t \end{bmatrix}.$$

However, the vector  $\mathbf{b}$  has been corrupted by noise and is not in the range of  $A$ .

- (a) (3 points) Find the singular values of  $A$ .
- (b) (4 points) When is  $\mathbf{b}$  not in  $\text{Ran}(A)$ ? When  $\mathbf{b} \notin \text{Ran}(A)$ , find the QR decomposition of the  $3 \times 3$  matrix  $[A \ \mathbf{b}]$  when  $t > 4$ . (I picked  $t > 4$  for no particular reason. It will streamline the process to make the diagonal entries of  $R$  positive.)
- (c) (2 points) Find the orthogonal projection matrix  $P$  onto the subspace  $\text{Ran}(A)$ .
- (d) (2 points) Find the least squares solution  $\hat{\mathbf{x}}$  that minimizes  $\|A\mathbf{x} - \mathbf{b}\|^2$ .
- (e) (3 points) (HARD) Let  $S$  be the unit circle in  $\mathbb{R}^2$ . Let  $B = \begin{bmatrix} 2 & 2 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$ . The image  $B(S)$  is an ellipse in  $\mathbb{R}^3$ . Find the lengths of the long and short half-axes of this ellipse, and find the vectors in  $\mathbb{R}^3$  that point along these axes.

**Answer:**1. SVD:

First, calculate  $A^T A = \begin{bmatrix} 3 & 2 \\ 2 & 2 \end{bmatrix}$ . The eigenvalues of  $A^T A$  are found by  $\det(A^T A - \lambda I) = \lambda^2 - 5\lambda + 2 = 0$ .

$$\lambda_{1,2} = \frac{5 \pm \sqrt{17}}{2}.$$

The singular values are  $\sigma_1 = \sqrt{\frac{5+\sqrt{17}}{2}}$  and  $\sigma_2 = \sqrt{\frac{5-\sqrt{17}}{2}}$ .

2. Range Condition and QR:

$\mathbf{b}$  is in the range if it is a linear combination of columns of  $A$ . Gaussian elimination shows that we need  $t \neq 3$  to avoid a contradiction.

**QR Decomposition:** Apply Gram-Schmidt to  $\mathbf{c}_1, \mathbf{c}_2, \mathbf{b}$ . (Remember that  $t > 4$ , so  $3 - t$  is negative.)

$$\begin{aligned} [A \quad \mathbf{b}] &= \begin{bmatrix} 1 & 1/3 & 3 \\ 1 & -2/3 & 2 \\ 1 & 1/3 & t \end{bmatrix} \begin{bmatrix} 1 & 2/3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \underbrace{\begin{bmatrix} 1 & 1/3 & \frac{3-t}{2} \\ 1 & -2/3 & 0 \\ 1 & 1/3 & \frac{t-3}{2} \end{bmatrix}}_Q \underbrace{\begin{bmatrix} 1 & 2/3 & \frac{t+5}{3} \\ 0 & 1 & \frac{t-1}{2} \\ 0 & 0 & 1 \end{bmatrix}}_R \\ &= \underbrace{\begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \end{bmatrix}}_Q \underbrace{\begin{bmatrix} \sqrt{3} & \frac{2}{\sqrt{3}} & \frac{t+5}{\sqrt{3}} \\ 0 & \frac{\sqrt{6}}{3} & \frac{t-1}{\sqrt{6}} \\ 0 & 0 & \frac{t-3}{\sqrt{2}} \end{bmatrix}}_R \end{aligned}$$

3. Projection Matrix:

Using  $P = I - \mathbf{u}_3 \mathbf{u}_3^T$  (where  $\mathbf{u}_3$  spans the orthogonal complement of  $\text{Range}(A)$ ):

$$\mathbf{u}_3 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}.$$

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} = I - \begin{bmatrix} 0.5 & 0 & -0.5 \\ 0 & 0 & 0 \\ -0.5 & 0 & 0.5 \end{bmatrix} = \begin{bmatrix} 0.5 & 0 & 0.5 \\ 0 & 1 & 0 \\ 0.5 & 0 & 0.5 \end{bmatrix}.$$

4. Least Squares Solution:

Solve normal equations  $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$ .

$$\begin{bmatrix} 3 & 2 \\ 2 & 2 \end{bmatrix} \hat{\mathbf{x}} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ t \end{bmatrix} = \begin{bmatrix} 5+t \\ 3+t \end{bmatrix}.$$

Inverse of  $\begin{bmatrix} 3 & 2 \\ 2 & 2 \end{bmatrix}$  is  $\frac{1}{2} \begin{bmatrix} 2 & -2 \\ -2 & 3 \end{bmatrix}$ .

$$\hat{\mathbf{x}} = \frac{1}{2} \begin{bmatrix} 2 & -2 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} 5+t \\ 3+t \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 10+2t-(6+2t) \\ -10-2t+9+3t \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 4 \\ t-1 \end{bmatrix} = \begin{bmatrix} 2 \\ \frac{t-1}{2} \end{bmatrix}.$$

5. Geometry of the Ellipse:

$$B = \begin{bmatrix} 2 & 2 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} = \underbrace{\begin{bmatrix} \frac{4}{3\sqrt{2}} & 0 & -\frac{1}{3} \\ \frac{1}{3\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{2}{3} \\ \frac{1}{3\sqrt{2}} & -\frac{1}{\sqrt{2}} & \frac{2}{3} \end{bmatrix}}_U \underbrace{\begin{bmatrix} 3 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}}_{\Sigma} \underbrace{\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}}_{V^T}$$

The first two columns of  $U$  are the long and short half-axis directions, and the singular values 3, 1 are the lengths.

5. Let  $A$  be a  $2 \times 2$  real matrix. We construct a larger  $4 \times 4$  block matrix  $K$  using polynomials of  $A$ :

$$K = \begin{bmatrix} A & A + I \\ A + I & A + 2I \end{bmatrix}.$$

- (a) (2 points) Calculate the determinant of  $K$ . (Hint: Use block Gaussian elimination to simplify the matrix structure).

- (b) (3 points) Find the inverse matrix  $K^{-1}$ . Express your answer in block form in terms of  $A$  and  $I$ .

- (c) (3 points) Let  $\mathbf{v}$  be an eigenvector of  $A$  with eigenvalue  $\lambda$  (assume  $\lambda \neq -1$ ). Find all scalars  $\alpha$  such that the vector  $\mathbf{w} = \begin{bmatrix} \mathbf{v} \\ \alpha \mathbf{v} \end{bmatrix}$  is an eigenvector of  $K$ .

- (d) (2 points) Now let  $A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ . Find all 4 eigenvalues of  $K$ .

- (e) (2 points) Under what condition of  $A$ , will the matrix  $K$  be positive definite?

**Answer:**

1. Determinant: Perform block row operation  $R_2 \leftarrow R_2 - R_1$ .

$$\det \begin{bmatrix} A & A+I \\ A+I & A+2I \end{bmatrix} = \det \begin{bmatrix} A & A+I \\ I & I \end{bmatrix}.$$

Since blocks commute (all are polynomials of  $A$ ), we can use  $\det(AD - BC)$ :  
 $\det(A(I) - (A+I)I) = \det(A - A - I) = \det(-I)$ . Since  $A$  is  $2 \times 2$ ,  $I$  is  $2 \times 2$ ,  
 so  $\det(-I_2) = (-1)^2 = 1$ .

$$\det(K) = 1.$$

2. Inverse: We solve  $\begin{bmatrix} A & A+I \\ A+I & A+2I \end{bmatrix} \begin{bmatrix} X & Y \\ Z & W \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}$ . Using row operations:  
 $\left[ \begin{array}{cc|cc} A & A+I & I & 0 \\ I & I & -I & I \end{array} \right]$  (after  $R_2 - R_1$ )  $\xrightarrow{R_1 - AR_2} \left[ \begin{array}{cc|cc} 0 & I & I+A & -A \\ I & I & -I & I \end{array} \right]$ . Then  
 we have  $\begin{bmatrix} 0 & I & I+A & -A \\ I & 0 & -2I-A & I+A \end{bmatrix}$ . Block swap gives:

$$K^{-1} = \begin{bmatrix} -(A+2I) & A+I \\ A+I & -A \end{bmatrix}.$$

3. Eigenvector Condition:  
 We compute  $K\mathbf{w}$ :

$$\begin{bmatrix} A & A+I \\ A+I & A+2I \end{bmatrix} \begin{bmatrix} \mathbf{v} \\ \alpha\mathbf{v} \end{bmatrix} = \begin{bmatrix} \lambda\mathbf{v} + \alpha(\lambda+1)\mathbf{v} \\ (\lambda+1)\mathbf{v} + \alpha(\lambda+2)\mathbf{v} \end{bmatrix} = \mu \begin{bmatrix} \mathbf{v} \\ \alpha\mathbf{v} \end{bmatrix}.$$

This gives the system for eigenvalue  $\mu$ :  $\lambda + \alpha(\lambda+1) = \mu$  and  $(\lambda+1) + \alpha(\lambda+2) = \mu\alpha$ . Substitute  $\mu$ :  $(\lambda+1) + \alpha(\lambda+2) = \alpha(\lambda + \alpha(\lambda+1)) = \alpha\lambda + \alpha^2(\lambda+1)$ .  
 Rearranging:

$$(\lambda+1)\alpha^2 - 2\alpha - (\lambda+1) = 0.$$

Solving for  $\alpha$ :

$$\alpha = \frac{2 \pm \sqrt{4 + 4(\lambda+1)^2}}{2(\lambda+1)} = \frac{1 \pm \sqrt{1 + (\lambda+1)^2}}{\lambda+1}.$$

4. Calculation:

Eigenvalues of  $A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$  are  $\lambda \in \{1, 0\}$ . Using  $\mu = \lambda + \alpha(\lambda+1)$ :

- Case  $\lambda = 0$ :  $\alpha^2 - 2\alpha - 1 = 0 \implies \alpha = 1 \pm \sqrt{2}$ .  $\mu = 0 + (1 \pm \sqrt{2})(1) = 1 \pm \sqrt{2}$ .
- Case  $\lambda = 1$ :  $2\alpha^2 - 2\alpha - 2 = 0 \implies \alpha^2 - \alpha - 1 = 0 \implies \alpha = \frac{1 \pm \sqrt{5}}{2}$ .  
 $\mu = 1 + \frac{1 \pm \sqrt{5}}{2}(2) = 1 + 1 \pm \sqrt{5} = 2 \pm \sqrt{5}$ .

5. Positive Definite Condition:

For  $K$  to be positive definite, all its eigenvalues must be positive. The eigenvalues of  $K$  are given by  $\mu = \lambda + 1 \pm \sqrt{1 + (\lambda + 1)^2}$ . Consider the eigenvalue corresponding to the negative root:  $\mu_- = (\lambda + 1) - \sqrt{(\lambda + 1)^2 + 1}$ . For any real  $\lambda$ ,  $\sqrt{(\lambda + 1)^2 + 1} > |\lambda + 1|$ . Therefore, if  $\lambda + 1 \geq 0$ ,  $\mu_- < 0$ . If  $\lambda + 1 < 0$ ,  $\mu_- < 0$ . Thus,  $K$  always has negative eigenvalues. **Conclusion:** There is NO condition on  $A$  (as a real matrix) that makes  $K$  positive definite. It is never positive definite.

6. Let  $V$  be the vector space of all  $2 \times 2$  real matrices with trace equal to zero. We consider the linear transformation  $L : V \rightarrow V$  defined by the commutator with a nilpotent matrix:

$$L(X) = JX - XJ, \quad \text{where } J = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Let  $\mathcal{B} = (E, H, F)$  be the ordered basis for  $V$ , where:

$$E = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad H = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad F = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}.$$

- (a) (3 points) Find the matrix representation of  $L$  under the basis  $\mathcal{B}$ .
- (b) (3 points) Find all eigenvalues of  $L$  along with their algebraic and geometric multiplicities. Is  $L$  diagonalizable?
- (c) (3 points) Consider a new ordered basis  $\mathcal{C} = (E', H', F')$  where  $E' = E$ ,  $H' = E + H$ , and  $F' = E + H + F$ . Find the change of coordinate matrix  $P_{\mathcal{B} \leftarrow \mathcal{C}}$  from  $\mathcal{C}$  to  $\mathcal{B}$ . Then, using this matrix, calculate the matrix representation of  $L$  under the new basis  $\mathcal{C}$ .



**Answer:**1. Matrix Representation ( $[L]_{\mathcal{B}}$ ):

We calculate the image of each basis vector:

- $L(E) = JE - EJ = \mathbf{0} - \mathbf{0} = 0 = 0E + 0H + 0F.$
- $L(H) = JH - HJ = \begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -2 \\ 0 & 0 \end{bmatrix} = -2E.$
- $L(F) = JF - FJ = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} - \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = 1H.$

Writing these coordinate vectors as columns:

$$A = [L]_{\mathcal{B} \leftarrow \mathcal{B}} = \begin{bmatrix} 0 & -2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

## 2. Eigenvalues and Diagonalizability:

Since  $A$  is upper triangular, the eigenvalues are the diagonal entries: **Eigenvalues:**  $\lambda = 0$  (Algebraic Multiplicity = 3).

To find the Geometric Multiplicity, we analyze  $\dim(\text{Null}(A - 0I)) = \dim(\text{Null}(A)).$

The matrix  $A$  has pivots in columns 2 and 3, so  $\text{Rank}(A) = 2$ . Geometric Multiplicity =  $\dim(V) - \text{Rank} = 3 - 2 = 1$ .

Since the Geometric Multiplicity (1) is strictly less than the Algebraic Multiplicity (3),  $L$  is **not diagonalizable**.

## 3. Change of Basis:

The change of coordinate matrix  $P = P_{\mathcal{B} \leftarrow \mathcal{C}}$  has columns equal to the coordi-

$$\begin{aligned} \text{nates of the } \mathcal{C} \text{ vectors expressed in } \mathcal{B}. \quad E' = 1E + 0H + 0F &\implies [E']_{\mathcal{B}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}. \\ H' = 1E + 1H + 0F &\implies [H']_{\mathcal{B}} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}. \quad F' = 1E + 1H + 1F \implies [F']_{\mathcal{B}} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}. \end{aligned}$$

$$P = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

To find  $[L]_{\mathcal{C} \leftarrow \mathcal{C}}$ , we compute  $P^{-1}AP$ . First, find  $P^{-1}$  (inverse of the upper triangular matrix):

$$P^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Now multiply:

$$AP = \begin{bmatrix} 0 & -2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -2 & -2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

$$[L]_{\mathcal{C} \leftarrow \mathcal{C}} = P^{-1}(AP) = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -2 & -2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -2 & -3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$